PDFlib PLOP: PDF Linearization, Optimization, Protection

Page inserted by evaluation version
www.pdflib.com – sales@pdflib.com

Renormalization of Self-Organized Critical Models

JEFF HASTY AND KURT WIESENFELD

School of Physics, Georgia Institute of Technology, Atlanta, Georgia 30332

ABSTRACT: Self-organized criticality refers to a mechanism whereby complex dissipative dynamical systems naturally evolve to a self-sustaining critical point, exhibiting correlations on all length scales and time scales. A recent theory based on a real space renormalization group approach sheds light on fundamental aspects of the phenomenon.

1. SELF-ORGANIZED CRITICALITY

Many phenomena can be explained theoretically by isolating a specific scale of interaction. This characteristic length or time scale is of great utility in that one can ignore or treat perturbative effects which are thought to occur on other scales. For example, the nuclear interaction can be ignored if one seeks to explain some atomic phenomenon such as chemical bonding. Thus one can speak of correlations which become vanishingly small and are therefore negligible. In dynamical systems, *scale invariance* refers to the antithesis of this picture. Scale invariant phenomena have no characteristic scale of interaction; small perturbations can propagate throughout the entire system and cannot be ignored.

The physical motivation for self-organized criticality (SOC) lies in the belief that there is a large class of phenomena where scale invariance arises naturally; that is, the underlying dynamical process leads to scale invariant observables. Couched in the language of equilibrium systems, the critical state is obtained without the fine-tuning of a parameter.

The prototypical model for SOC is known as the BTW sandpile model, which was introduced nearly a decade ago in the first work on the subject [1]. On an $L \times L$ lattice, the cellular automaton (CA) model is defined by two simple updating rules. First the pile is driven by randomly choosing a site (i, j) and dropping a grain of "sand" by updating the height h_{ij},

$$h_{i,j} \to h_{i,j} + 1. \tag{1. 1}$$

Secondly, if the local height exceeds the threshold value h_c, the site relaxes by distributing sand to its nearest neighbors,

$$\begin{aligned} h_{i,j} &\to h_{i,j} - 4 \\ h_{i\pm1,j} &\to h_{i\pm1,j} + 1 \\ h_{i,j\pm1} &\to h_{i,j\pm1} + 1. \end{aligned} \tag{1.2}$$

17496632, 1998, 1, Downloaded from https://nyaspubs.onlinelibrary.wiley.com/doi/10.1111/j.1749-6632.1998.tb09853.x by University Of California, Wiley Online Library on [02/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

This relaxation can then cause neighboring sites to become excited (above threshold), thus triggering a sequence of relaxation events known as an avalanche. A separation of time scales is built into the simulation; dropping does not take place until all sites are relaxed. Sand leaves the system by "falling off" at the boundary.

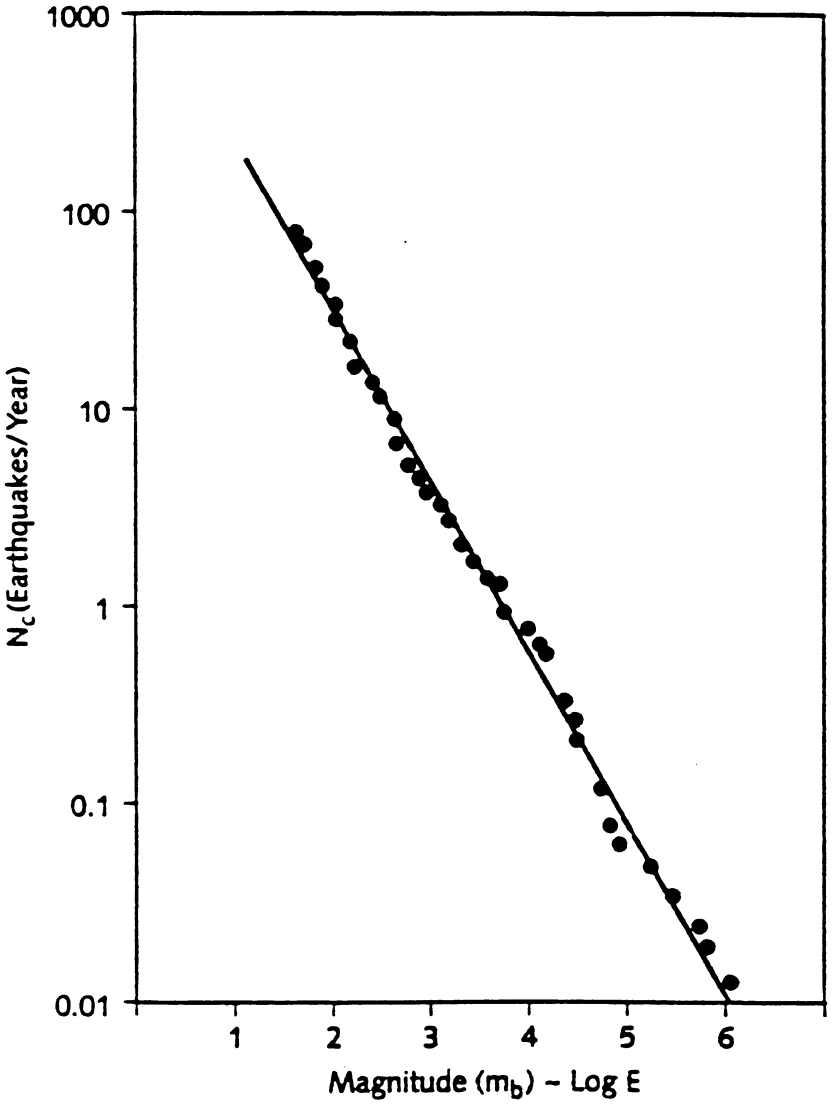


FIGURE 1. Earthquake size distribution along the New Madrid fault zone from 1973–1983 (reproduced from [5]).

17496632, 1998, 1, Downloaded from https://nyaspubs.onlinelibrary.wiley.com/doi/10.1111/j.1749-6632.1998.tb09853.x by University Of California, Wiley Online Library on [02/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

This model, which is intended more as a paradigm than a real simulation of sand, provides a framework for studying certain driven-diffusive systems. Independent of initial conditions, the simulations evolve through a transitory period to a state characterized by power laws in the statistical distributions of certain observables. For example, the avalanche size is defined as the number of sites which relax in a single avalanche, with distribution given by

$$D(s) \sim s^{1-\tau}, \tag{1. 3}$$

where $D(s)$ is the probability that s or more sites are involved.

There are of course numerous spatial and temporal measurements which can be made, and there has been considerable work in the direction of deducing various scaling laws which relate their distribution exponents [2]. Here, however, we will maintain a narrow focus, concentrating solely on the avalache size distribution.

Self-organized criticality has been invoked to explain many diverse phenomena, with earthquakes being perhaps the most widely cited example [3]. An empirical result describing the distribution of earthquake sizes is the well established Gutenberg-Richter law [4]. In FIG. 1, the earthquake size distribution obtained from 1973–1983 along the New Madrid fault zone in the central United States is plotted [5]. The distribution obeys $E(m) \sim m^{-\beta}$, where $E(m)$ is the number of earthquakes of size m or greater. That such a simple fit would arise is remarkable, but perhaps even more astounding is the claim that this distribution is obeyed in any geographical region, with the only region-specific quantity being the value of the exponent β.

A simple model used by geophysicists to describe one class of earthquakes is that of two plates shearing against one another. In this picture, plate tectonics slowly drives one plate against another until some local position exceeds a static friction threshold and slips. This slipping can then cause neighboring areas to slip, and an avalanche of slipping events is an earthquake. In 1967 geophysicists Burridge and Knopoff [6] proposed a spring-block model for this type of plate dynamics. In their model, a fixed plate supports a two-dimensional array of blocks connected by springs to each other and to a moving plate (see FIG. 2.).

In 1992 Chirstensen and Olami [7] analyzed the Burridge-Knopoff model with an eye towards SOC. They were able to show how a dissipative sandpile model can be derived from the equations of motion by using a threshold description of the friction-

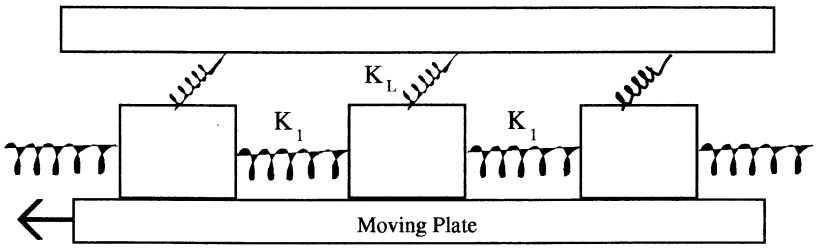


FIGURE 2. Schematic of the Burridge-Knopoff spring-block model.

17496632, 1998, 1, Downloaded from https://nyaspubs.onlinelibrary.wiley.com/doi/10.1111/j.1749-6632.1998.tb09953.x by University Of California, Wiley Online Library on [02/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

al force and a separation of time scales. For example, if one lets the displacement of each block in FIG. 2 be denoted $x_{i,j}$, then the total force on block (i, j) is given by

$$F_{i,j} = K_1(2x_{i,j} - x_{i-1,j} - x_{i+1,j}) + K_2(2x_{i,j} - x_{i,j-1} - x_{i,j+1}) + K_L x_{i,j}, \qquad (1.4)$$

where the K's denote the elastic constants for the springs. Now the general idea is to let the plates slide slowly against each other until one site reaches some threshold and slips, triggering the process of relaxation. Christensen and Olami show that the redistribution of force after a local slip at position $x_{i,j}$ is given by

$$\begin{aligned} F_{i\pm1,j} &\to F_{i\pm1,j} + \delta F_{i\pm1,j} \\ F_{i,j\pm1} &\to F_{i,j\pm1} + \delta F_{i,j\pm1} \\ F_{i,j} &\to 0, \end{aligned} \qquad (1.5)$$

where the increase in the force on the nearest-neighbor blocks is

$$\begin{aligned} \delta F_{i\pm1,j} &= \frac{K_1}{2K_1 + 2K_2 + K_L} F_{i,j} = \alpha_1 F_{i,j} \\ \delta F_{i,j\pm1} &= \frac{K_2}{2K_1 + 2K_2 + K_L} F_{i,j} = \alpha_2 F_{i,j}. \end{aligned} \qquad (1.6)$$

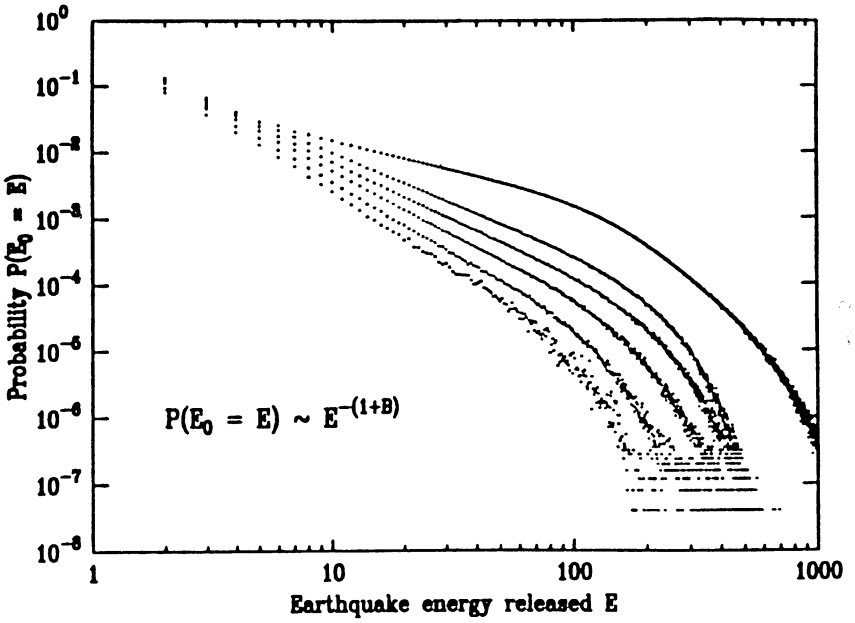


FIGURE 3. Simulation results of the Christensen-Olami model (reproduced from [7]). Plotted is the distribution for the number of slips in an avalanche (energy released in an earthquake), with the differing curves representing differing amounts of dissipation.

17496632, 1998, 1, Downloaded from https://nyaspubs.onlinelibrary.wiley.com/doi/10.1111/j.1749-6632.1998.tb09853.x by University Of California, Wiley Online Library on [02/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Next an algorithm for simulating the system is generated by exploiting a supposed separation of time scales; one assumes that in real systems the relaxation process (earthquakes) are orders of magnitude faster than the driving (plate motion), so that the system need only be driven when all sites are stable. The simulation can then be summarized as follows: A random site is chosen and a small amount of force is added. When the force at a given site exceeds some threshold value F_{th}, this site will relax according to (1.5). This in turn can cause the neighboring sites to exceed F_{th} so that they then may relax to their neighbors, and so on. This relaxation event or earthquake continues until no sites exceed F_{th}, whence the driving continues by choosing a site randomly as before.

This model is very similar to the BTW model previously described, with the notable difference being the possible lack of local conservation; for $\alpha_1 = \alpha_2 = .25$ the model is locally conservative, but for other values of α the model is not. The simulation results reproduce the Guttenburg-Richter law, yielding differing exponents depending on the degree of local conservation (see FIG. 3). This result is quite interesting in light of empirical data which suggests that the exponential value is a function of the media, and one could argue heuristically that the K values (and hence the α values) should depend on the geographic region.

2. A RENORMALIZATION THEORY FOR SOC

Although CA models such as those explained above are easy to simulate, a general framework for deducing statistical distributions from them has not been fully developed. In this section we explain a recently proposed renormalization scheme capable of calculating distribution exponents and explaining why different models can exhibit the same exponent values (universality).

The Pietronero, Vespignani, Zapperi (PVZ) [8] approach uses a real space coarse-graining of the local dynamics to yield a renormalization group (RG) procedure. The original implementation was in the context of the BTW model described above and a similiar model known as the Manna model [9], [10]. The Manna model differs from the BTW model only in its relaxation rule; an excited site chooses two neighbors at random for distribution. Both models evolve to a stationary state characterized by power law distributions for various observables. In the SOC state, the Manna simulation yields an avalanche exponent of $\tau = 1.28$ [10], close to the value of 1.22 obtained for the BTW model [1].

The first step in the PVZ theory is to define a space of relaxation probabilities, whereby each point in the space corresponds to a specific model. This space is spanned by the four probabilities P_j ($j = 1, 2, 3, 4$), with a point in the space described by the vector $\mathbf{P} = (P_1, P_2, P_3, P_4)$. Within this framework, the BTW and Manna models correspond to $\boldsymbol{P} = (0, 0, 0, 1)$ and $P = (0, 1, 0, 0)$, since the probability that the BTW model relaxes by distributing sand to four neighbors is unity, while the Manna model relaxes to two neighbors with probability one.

The next step is to calculate how a given point evolves under a change of scale. PVZ first define a 2×2 block of sites with one site excited initially (above threshhold), and then find all relaxation paths generated from the excited block. How each path affects its neighboring blocks is monitored, and a sum is taken over the paths

17496632, 1998, 1, Downloaded from https://nyaspubs.onlinelibrary.wiley.com/doi/10.1111/j.1749-6632.1998.tb09853.x by University Of California, Wiley Online Library on [02/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

leading to a given block relaxation. This is done for all possible initial block configurations (a block configuration being composed of internal threshhold and below-threshhold sites), and a weighted average is taken. The weight factor is determined by the probability of obtaining a given initial block, and this involves an additional parameter ρ which defines the density of threshhold sites. ρ is itself determined from **P** via the *balance condition*,

$$1 = \rho(P_1 + 2P_2 + 3P_3 + 4P_4), \tag{2.1}$$

which effectively defines ρ so that the local conservation rule (sand in = sand out) is satisfied on every scale.

This procedure yields a map of the form $\mathbf{P}'(\mathbf{P})$, where $\mathbf{P}'$ denotes the block relaxation probabilities. The map is then iterated with either the BTW or the Manna initial conditions, and a unique fixed point common to both models is found. This fixed point determines the exponent τ (cf. Eq. (1.3)), and PVZ obtain a value of $\tau = 1.253$, in good agreement with the simulation results cited above.

3. RENORMALIZATION OF A DIRECTED SANDPILE MODEL

It is impressive that this approach generates the right exponent τ from first principles. However, the RG picture predicts much more, and its success or failure should rest on a systematic check of all of its predictions. In this section we explain the application of a PVZ-type RG scheme to a model known as the directed sandpile model [11]. This model, while not as complex as the BTW or Manna models, enables us to explore the implications of the RG theory, and to test its validity. Further, there are known analytic results [11] for the avalanche exponents.

The directed model is defined as follows. The system is perturbed by randomly choosing one of the sites on the upper row of the lattice and updating;

$$h_{1,j} \to h_{1,j} + 1. \tag{3.1}$$

Next, if the height h at any site exceeds 1, then the site relaxes by transferring one grain to each of its two forward neighbors:

$$\begin{aligned} h_{i,j} &\to h_{i,j} - 2 \\ h_{i+1,j\pm} &\to h_{i+1,j\pm} + 1, \end{aligned} \tag{3.2}$$

where $j\pm = j \pm [1 \pm (-1)^i]/2$. We will refer to this model as the DR model.

Dhar and Ramaswamy [11] have shown analytically that, in the scaling region, the avalanche distribution is given by $P(s) \sim s^{-4/3}$, where $P(s)$ is the probability that s sites tumble in a single event.

As before, the RG scheme specifies how the rule (3.2) transforms under a change of scale. Mathematically, it provides a mapping on the space where each point represents a particular CA rule.

The explicit application involves the summing of all internal relaxation paths which lead to a given type of diamond relaxation, and the details will be provided elsewhere. The end result is that an RG operator is derived which transforms the dy-

17496632, 1998, 1, Downloaded from https://nyaspubs.onlinelibrary.wiley.com/doi/10.1111/j.1749-6632.1998.tb09853.x by University Of California, Wiley Online Library on [02/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

namics of inter-diamond sites (which are themselves diamonds) to a dynamics for the coarse-grained diamond cell. The space on which the RG operator acts is thus a

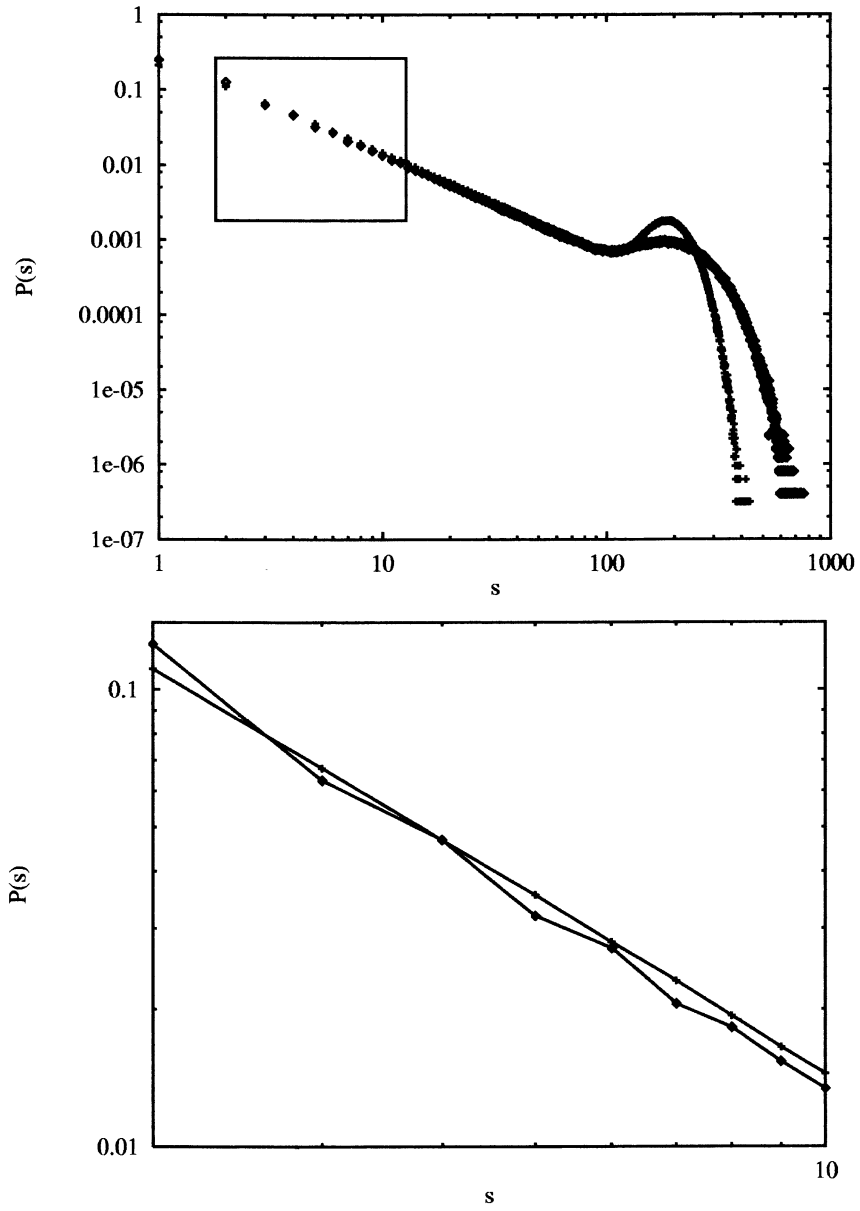


FIGURE 4. Avalanche distribution for the model corresponding to the fixed point of the RG map (+) and the original DR model data (o). The small length scale blow up highlights the power law behavior of the fixed point model.

17496632, 1998, 1, Downloaded from https://nyaspubs.onlinelibrary.wiley.com/doi/10.1111/j.1749-6632.1998.tb09853.x by University Of California, Wiley Online Library on [02/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

space where each point represents a particular cellular automata rule, with the directed model being one example.

The RG map can then be analyzed, and the results checked against the analytic results or by using simulations. For the initial condition rule corresponding to the DR model, the RG map is attracted to a fixed point. This fixed point implies that there is indeed scale invariance, and we can calculate the avalanche exponent from it. We obtain $\tau = 1.325$, in excellent agreement with the known result. We next simulate the rule corresponding to this fixed point and measure the avalanche distribution (FIG. 4). This data gives an exponent of $\tau = 1.32$, in the same class as the directed model as predicted by the theory. Further, the small length scale power law deviations in the directed model data have disappeared, also as predicted by the RG picture: the DR model does not correspond to a fixed point and therefore we do not expect scale invariance at small scales, while the fixed point model should be invariant at all scales.

The RG theory predicts the existence of universality classes corresponding to the basin of attraction of each fixed point; all models in the basin should share a common avalanche exponent. We find that the above-mentioned fixed point has a large basin of "symmetric rules" described as follows. If we let R/L be the probability that a site relaxes by distributing sand to its lower right/left neighbors, then symmetric rules are those which have $R = L$. Preliminary simulation results for the avalanche distributions for various symmetric rules yield a common exponent for all models as predicted. Additionally, we find that the further a rule is from the fixed point (as measured by the number of iterates), the larger the region of small avalanche power law deviation, again consistent with the RG picture.

4. SUMMARY

The RG implementation presented above provides valuable information about SOC models. Further generalizations could yield still more. For example, the size of the system could be included in the analysis, potentially explaining finite size effects such as the exodus from the scaling region in data such as FIG. 4. Another unexplored area is the transient dynamics as the system evolves toward the SOC state, and it may be possible to learn something of the self-organizing behavior by exploiting the RG method.

There are also limitations to the current RG approach as it is currently applied. In particular, coarse-graining inevitably throws out correlations inherent on the attractor [12], and there is as yet no systematic method for including higher order corrections to the RG map. Potentially, these are only technical points provided the overall structure of the theory hangs together, as our work on the directed models indicates. Another open question is whether a complete RG theory will be able to explain the nonuniversal behavior seen in the Christensen-Olami model.

We have described one way in which cellular automata are used to model nonequilibrium phenomena, and how SOC can arise as a consequence. In modeling such "complex systems," the specification of all relevant details yields models which are often beyond analysis. Universality, in this respect, is a trump card. If one can demonstrate that a large number of models have common features, then one may proceed

17496632, 1998, 1, Downloaded from https://nyaspubs.onlinelibrary.wiley.com/doi/10.1111/j.1749-6632.1998.tb09853.x by University Of California, Wiley Online Library on [02/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

by analyzing the simplest of the models. The power of a renormalization theory lies not only in its ability to deduce critical exponents, but also in its explanation and prediction of such universality classes.

REFERENCES

1. P. BAK, C. TANG & K. WIESENFELD. 1987. Self-organized criticality: An explanation of 1/f noise. Phys. Rev. Lett. **59:** 381–384; 1988. Self-organized criticality. Phys. Rev. A **38:** 364–374.
2. C. TANG & P. BAK. 1988. Mean field theory of self-organized critical phenomena. J. Stat. Phys. **51:** 797–802; K. CHRISTENSEN, H.C. FOGEDBY & H.J. JENSEN. 1991. Dynamical and spatial aspects of sandpile cellular automata. J. Stat. Phys. **63:** 653–684.
3. P. BAK & C. TANG. 1989. Earthquakes as a self-organized critical phenomenon. J. Geophys. Res. B. **94:** 15635–15637; A. SORNETTE & D. SORNETTE. 1989. Self-organized criticality and earthquakes. Europhys. Lett. **9:** 197–202; K. CHEN, P. BAK & S.P. OBUKHOV. 1990. Self-organized criticality in a crack-propagation model of earthquakes. Phys. Rev. A. **43:** 625–630; K. CHRISTENSEN & Z. OLAMI. 1992. Variation of the Gutenberg-Richter *b* values and nontrivial temporal correlations in a spring-block model for earthquakes. J. Geophys. Res **97:** 8729–8735.
4. B. GUTENBERG & C.F. RICHTER. 1956. Magnitude and energy of earthquakes. Ann. di Geofis. **57:** 1–15.
5. A.C. JOHNSTON & S. NAVA. 1985. Recurrence rates and probability estimates for the New Madrid seismic zone. J. of Geophys. Res. **90:** 6737–6753.
6. R. BURRIDGE & L. KNOPOFF. 1967. Model and theoretical seismicity. Bull. Seismol. Soc. Am. **57:** 341–371.
7. K. CHRISTENSEN & Z. OLAMI. 1992. Variation of the Gutenberg-Richter *b* values and nontrivial temporal correlations in a spring-block model for earthquakes. J. Geophys. Res **97:** 8729–8735.
8. L. PIETRONERO, A. VESPIGNANI & S. ZAPPERI. 1994. Renormalization scheme for self-organized criticality in sandpile models. Phys. Rev. Lett. **72:** 1690–1693; A. VESPIGNANI, S. ZAPPERI & L. PIETRONERO. 1995. Renormalization approach to the self-organized critical behavior of sandpile models. Phys. Rev. E. **51:** 1711–1724.
9. S.S. MANNA. 1991. Two-state model of self-organized critical phenomena. J. Phys. A **24:** L363–L369.
10. S.S. MANNA. 1990. Large-scale simulation of avalanche cluster distribution in sand pile model. J. Stat. Phys. **59:** 509–521; 1991. Critical exponents of the sandpile models in two dimensions. Physica A. **179:** 249–268.
11. D. DHAR & R. RAMASWAMY. 1989. Exactly solved model of self-organized critical phenomena. Phys. Rev. Lett. **63:** 1659–1662.
12. J. HASTY & K. WIESENFELD. 1997. Renormalization of one-dimensional avalanche models. J. Stat. Phys. **86:** 1179–1201.

17496632, 1998, 1, Downloaded from https://nyaspubs.onlinelibrary.wiley.com/doi/10.1111/j.1749-6632.1998.tb09853.x by University Of California, Wiley Online Library on [02/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License